

Flexible and Fully Quantized Ultra-Lightweight TinyissimoYOLO for Ultra-Low-Power Edge Systems

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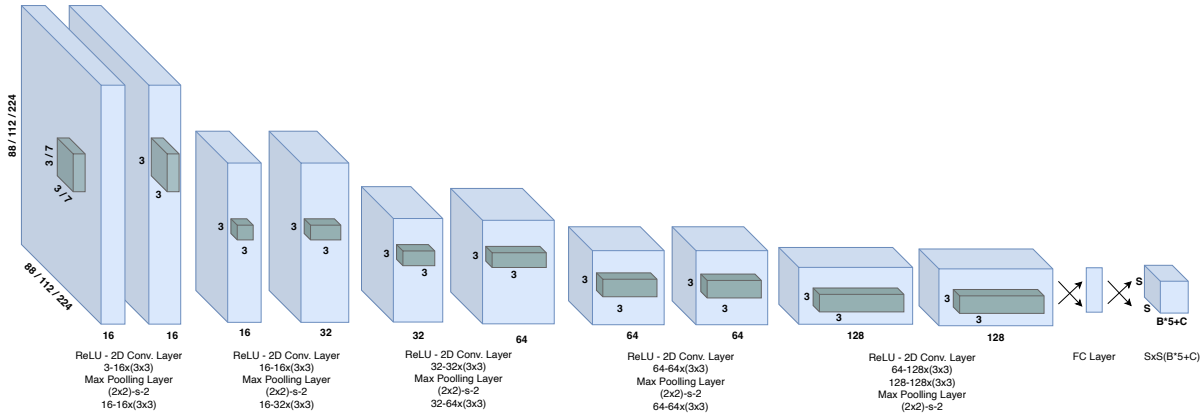


Figure 1: TinyissimoYOLO as deployed on GAP9 (both the 8-core cluster and the hardware accelerator) by this paper.

ABSTRACT

This paper deploys and explores variants of TinyissimoYOLO, a highly flexible and fully quantized ultra-lightweight object detection network designed for edge systems with a power envelope of a few milliwatts. With experimental measurements, we present a comprehensive characterization of the network’s detection performance, exploring the impact of various parameters, including input resolution, number of object classes, and hidden layer adjustments. We deploy variants of TinyissimoYOLO on state-of-the-art ultra-low-power extreme edge platforms, presenting an in-depth a comparison on latency, energy efficiency, and their ability to efficiently parallelize the workload. In particular, the paper presents a comparison between a novel RISC-V-based parallel processor (GAP9 from Greenwaves) with and without use of its on-chip hardware accelerator, an ARM Cortex-M7 core (STM32H7 from ST Microelectronics), two ARM Cortex-M4 cores (STM32L4 from STM and

Apollo4b from Ambiq), and a multi-core platform aimed at edge AI applications with a CNN hardware accelerator (Analog Devices MAX78000). Experimental results show that the GAP9’s hardware accelerator achieves the lowest inference latency and energy at 2.12 ms and 150 μ J respectively, which is around 2x faster and 20% more efficient than the next best platform, the MAX78000. The hardware accelerator of GAP9 can even run an increased resolution version of TinyissimoYOLO with 112×112 pixels and 10 detection classes within 3.2 ms, consuming 245 μ J. To showcase the competitiveness of a versatile general-purpose system we also deployed and profiled a multi-core implementation on GAP9 at different operating points, achieving 11.3 ms with the lowest-latency and 490 μ J with the most energy-efficient configuration. With this paper, we demonstrate the flexibility of TinyissimoYOLO and prove its detection accuracy detection datasets. We demonstrate its suitability for real-time ultra-low-power edge inference by benchmarking its performance.

*All three authors contributed equally to this research.

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KEYWORDS

YOLO, ML, computer vision, object detection, hardware accelerator, microcontroller, quantization, quantization-aware training, network deployment, network deployment evaluation